

Linhai Ma

Ph. D.

New Haven, Connecticut, United States • 9198840535

rinnkai.ba2019@gmail.com

He is a Postdoctoral Researcher at Yale University, where his work centers on natural language processing for healthcare, particularly extracting patient-centered information from secure clinical communications using pre-trained and large language models (LLMs).

Previously, he was a Machine Learning Engineer at Cadence Design Systems in California, developing AI infrastructure and applying LLMs in industrial settings.

His expertise includes three main areas:

- Deep learning: robustness and generalization of neural networks.
- Biomedical Image Processing: medical signal and image analysis (classification, segmentation, landmark detection, image generation, and translation).
- Natural language processing: named entity recognition, sentence classification, question answering, and topic modeling.

In addition, he has extensive industry-level experience in developing machine learning systems, AI algorithm design, and data processing.

Links

Page: <https://sarielma.github.io/>, Github: <https://github.com/sarielma>, LinkedIn: <https://www.linkedin.com/in/linhai-ma-ph-d-6441861bb/>

Education

Ph.D. in Computer Science

University of Miami, Coral Gables, FL – 2017 — 2023

Research interest: Robustness of deep neural networks

Master of Science in Computer Science

Institute of Software, Chinese Academy of Sciences, Beijing, China – 2014 — 2017

Research Interest: Concurrency

Bachelor of Science in Information Security

Northeastern University, Shenyang, China – 2010 — 2014

Courses: C/C++, Java, Data Structure, Computer Architecture, Operating System, Computer Network, etc.

Programming Skills

Programming Language:

Python, C/C++, Java, C#, Ruby, R, SQL, Prolog, etc.

Framework:

Pytorch, Pandas, Scikit-learn, Numpy, Kubernetes, Docker, PyQt5, Java Swing, etc.

Recent Industry Experiences

Cadence Design Systems, Inc., San Jose, CA
Machine Learning Engineer

Jul 2024 — Dec 2024

- Deployed large language models (LLMs) for EDA queries using NVIDIA NIMs, and configured URL routing through Kubernetes.
- Optimized retrieval-augmented generation (RAG) performance, reducing response latency from 10s to 100ms by profiling production code and rewriting vector database filtering logic.
- Improved document retrieval accuracy in RAG by fine-tuning the Hierarchical Navigable Small World (HNSW) algorithm.
- Collaborated with NVIDIA teams to refine and enhance LLM-based product performance.

Cadence Design Systems, Inc., San Jose, CA
Intern Software Engineer in Machine Learning

May 2022 — Aug 2022

- Designed and implemented a machine learning system to accelerate worst-case measurement evaluation of integrated circuits (ICs), reducing discovery time by over 90% compared to traditional simulation-based methods.
- Built Gaussian Process regression models in Python/PyTorch, trained on simulation data sampled via Latin Hypercube Sampling.
- Applied Bayesian optimization to iteratively select simulation inputs most likely to produce worst-case measurements, dynamically updating the training dataset.
- Performed forward feature selection to remove noisy input features and reduce simulation outliers.

Cadence Design Systems, Inc., San Jose, CA
Intern Software Engineer in Machine Learning

May 2021 — Aug 2021

- Designed a machine learning system to predict integrated circuit (IC) simulation outputs using Python and Scikit-learn.
- Built a data pipeline to preprocess historical simulation data into structured training sets.
- Implemented a Random Forest model to handle highly diverse simulation input ranges, achieving a mean absolute error of 0.99 picoseconds.
- Developed an interactive UI with PyQt5 and Matplotlib to visualize feature impact on simulation outcomes.

Hillstone Networks, Beijing, China
Intern Software Engineer (C/C++)

Jul 2015 — Nov 2015

- Migrated an Intrusion Prevention System (IPS) to new server hardware and updated the application stack using Ruby and Ruby on Rails.
- Implemented a packet sniffer in C/C++ to capture and process network traffic periodically for threat analysis.
- Automated backend SQLite operations with a tool chain using Python, SQL, and C/C++, enabling programmatic database updates in response to client requests.
- Implemented an HTTP URI filter in the firewall (C/C++) to detect and block abnormal or suspicious requests.

Huaxin Education Technology Co., Ltd., Shenyang, China
Intern Software Engineer (Java)

Jun 2013 — Aug 2013

- Developed a web application vulnerability scanner in Java/JSP with Oracle backend, capable of detecting common attacks such as SQL injection and cross-site scripting (XSS).

- Built a user interface in Java Swing to improve usability and streamline vulnerability reporting.

Related Research Experiences

Postdoctoral Associate at Yale University, New Haven, CT • Jan 2025 — Present

- Developing a pipeline to extract patient-centered information from secure clinical communications using pre-trained models (BERT variants) and large language models (LLMs).
- Enhancing LLM extraction performance through prompt engineering and instruction fine-tuning.
- Improving BERT-based models by incorporating context derived from BERTopic trained on unlabeled data.

Postdoctoral Associate at University of Florida, Gainesville, FL • Jan 2024 — Jun 2024

- Designed a CNN-based model for cervical vertebrae maturation (CVM) stage classification, achieving 76.7% accuracy.
- Developed diffusion models for translating medical images across different modalities.
- Built a multi-modal deep learning system to support prostate tumor diagnosis by integrating micro-ultrasound images with non-image clinical data.

Research Assistant at University of Miami, Coral Gables, FL • Sep 2019 — Dec 2023

- Developed a regularization method to reduce the impact of small adversarial noise on deep learning models; applied it to ECG signal data.
- Designed adaptive adversarial training algorithms with sample-specific margin adjustment, improving robustness against adversarial noise while maintaining model accuracy.
- Proposed a loss-defined margin algorithm that generalizes across medical imaging tasks, enhancing robustness in MR image segmentation, X-ray landmark detection, and blood cell object detection.

Research Assistant at University of Miami and Northwestern University, Coral Gables, FL • Jan 2018 — Aug 2019

- Identified evolution patterns of successful academic groups through large-scale data mining.
- Parsed author and publication records (1991–2018) from the Microsoft Graph database using SQL, and constructed million-scale annual coauthor networks with Python and NetworkX.
- Designed a Monte Carlo-based clustering algorithm in Python and R to detect author groups within networks.
- Defined group evolution patterns (e.g., splitting, merging) across adjacent years.
- Analyzed the correlation between group evolution and academic success (e.g., citation counts) using statistical methods in Python and R.

Research Assistant at Institute of Software, Chinese Academy of Sciences, Beijing, China • Sep 2015 — May 2017

- Developed a multi-threaded C++ test case generator for validating concurrent data structures.
- Proposed and implemented three adaptive generation strategies, increasing concurrency error detection by up to 6% while reducing execution time by up to 10%.

Selected Reviewer Experience

NeurIPS (2022-2025)

ICML (2022, 2025)

ICLR (2024-2026)

AISTATS (2026)

AAAI (2026)

AI

Neurocomputing

Teaching Experience

Teaching Assistant of CB&B 750 01 (SP25): Core Topics in Biomedical Informatics, 2025 Spring at Yale University

Teaching Assistant of CSC546 Introduction to Machine Learning, 2023 Spring at University of Miami

Teaching Assistant of CSC315 Introduction to Python for Scientist, 2023 Spring at University of Miami

Teaching Assistant of CSC546 Introduction to Machine Learning, 2022 Fall at University of Miami

Teaching Assistant of CSC315 Introduction to Python for Scientist, 2020 Spring at University of Miami

Teaching Assistant of CSC315 Introduction to Python for Scientist, 2019 Fall at University of Miami

Teaching Assistant of CSC546 Introduction to Machine Learning, 2019 Spring at University of Miami

Teaching Assistant of CSC421 Computer Operating Systems, 2018 Fall at University of Miami

Teaching Assistant of CSC421 Computer Operating Systems, 2017 Fall at University of Miami

Publications

Intrathread Method Orders Based Adaptive Testing of Concurrent Objects • 2025

Y Dai, L Xie, P Wu, S Cui, L Ma

Science of Computer Programming, 103362

Attention-based Shape-Deformation Networks for Artifact-Free Geometry Reconstruction of Lumbar Spine from MR Images • 2025

L Qian, J Chen, L Ma, T Urakov, W Gu, L Liang

IEEE transactions on medical imaging

A novel attention-based network for geometry reconstruction with error estimation from medical images • 2025

L Qian, J Chen, L Ma, T Urakov, W Gu, L Liang

Medical Imaging 2025: Image Processing 13406, 169-176

Symtc: A symbiotic transformer-cnn net for instance segmentation of lumbar spine MRI • 2024

J Chen, L Qian, L Ma, T Urakov, W Gu, L Liang

Computers in biology and medicine 179, 108795

Adaptive Testing of Concurrent Objects • 2024

Y Dai¹, P Wu, S Cui¹, L Ma

Theoretical Aspects of Software Engineering: 18th International Symposium, TASE 2024, Guiyang, China, July 29–August 1, 2024, Proceedings

Evaluation of Deep Neural Network Models for Instance Segmentation of Lumbar Spine MRI • 2024

J Chen, L Qian, L Ma, T Urakov, W Gu, L Liang

bioRxiv, 2024.04. 02.587810.

A sequential geometry reconstruction based deep learning approach to improve accuracy and consistence of lumbar spine MRI image segmentation • 2024

L Qian, J Chen, L Ma, T Urakov, L Liang

Medical Imaging 2024: Image Processing 12926, 753-760

A general approach to improve adversarial robustness of DNNs for medical image segmentation and detection. • 2024

L Ma, J Chen, L Qian, L Liang

Medical Imaging 2024: Image Processing 12926, 266-274

Towards Improving the Adversarial Robustness of Deep Neural Networks • 2023

L Ma

University of Miami

Improving Adversarial Robustness of Deep Neural Networks via Adaptive Margin Evolution. • 2023

L Ma, L Liang

Neurocomputing 551 (126524)

Increasing-Margin Adversarial (IMA) Training to Improve Adversarial Robustness of Neural Networks. • 2023

L Ma, L Liang

Computer Methods and Programs in Biomedicine 240 (107687)

Towards lifting the trade-off between accuracy and adversarial robustness of deep neural networks for medical image classification and segmentation. • 2023

L Ma, L Liang

Medical Imaging 2023: Image Processing 12464, 429-436

Adaptive Adversarial Training to Improve Adversarial Robustness of DNNs for Medical Image Segmentation and Detection. • 2023

L Ma, L Liang

<https://arxiv.org/pdf/2206.01736.pdf>

A Regularization Method to Improve Adversarial Robustness of Neural Networks for ECG Signal Classification. • 2022

L Ma, L Liang

Computers in Biology and Medicine 144 (105345)

Enhance CNN Robustness Against Noises for Classification of 12-Lead ECG with Variable Length. • 2020

Linhai Ma, Liang Liang

19th IEEE international conference on machine learning and applications (ICMLA 2020).

An Algorithm for Out-Of-Distribution Attack to Neural Network Encoder. • 2020

L Liang, L Ma, L Qian, J Chen

<https://arxiv.org/pdf/2009.08016.pdf>

Improve robustness of DNN for ECG signal classification: a noise-to-signal ratio perspective. • 2020

L Ma, L Liang

International Conference on Learning Representations (ICLR 2020) Workshop AI for Affordable Health.

Diversity driven adaptive test generation for concurrent data structures. • 2018

L Ma, P Wu, TY Chen

Information and software technology 103, 162-173